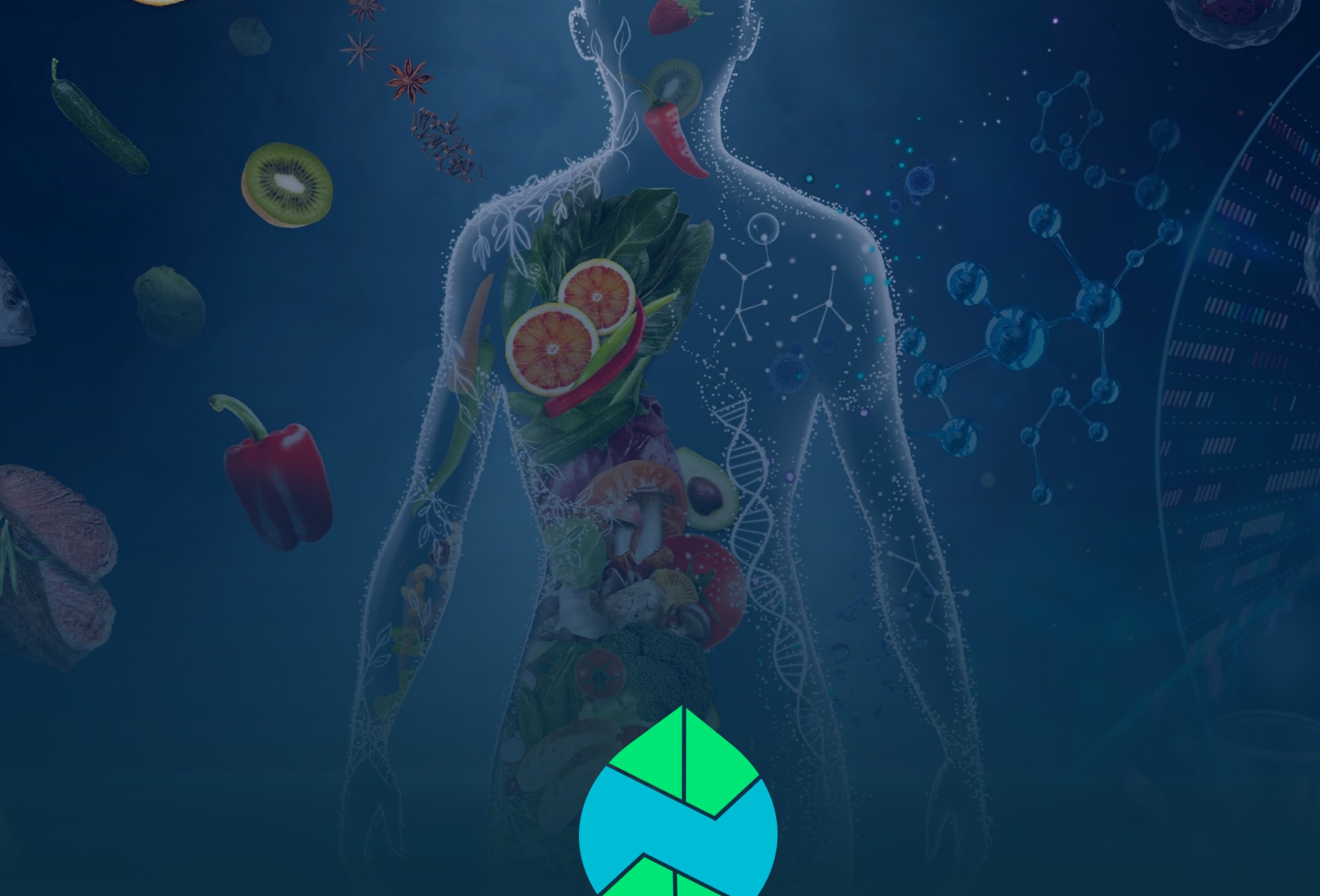

PROJECT DETAIL • SOFTWARE SPECIFICATION

# NutriSense

*AI-Powered Nutrition, Diet & Clinical Health Companion*

Technical Project Detail & Architecture Document

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# 1. Executive Summary & Project Introduction

NutriSense is an enterprise-grade, AI-powered nutritional intelligence, fasting management, and clinical health platform designed specifically for Pakistani and South Asian users. Engineered to combat the national surge in lifestyle-related metabolic disorders—principally Type 2 Diabetes, Hypertension, and Obesity—NutriSense makes personalized, culturally-accurate nutrition and fitness guidance accessible to every household, regardless of income.

## 1.1 Purpose & Problem Statement

**The Problem:** Pakistan faces a nutrition crisis – over 33% of adults are diabetic or pre-diabetic, with rising obesity, micronutrient deficiencies, and a high poverty rate that leaves most families unable to afford professional nutritionist consultations. Existing nutrition apps are built for Western diets: they ignore traditional portion units (katori, roti, naan), don't account for fasting practices like Ramadan, and are English-only – excluding the majority of users. Patients with diabetes or hypertension receive no personalized, clinically-safe dietary or exercise guidance, and families managing nutrition for children or elderly parents have no unified tool.

## 1.2 Target Audience

**Who it affects:** Underweight individuals, diabetics, people with high blood pressure, and those with obesity – the groups most vulnerable to preventable nutrition-related health complications.

## 1.3 Key Value Propositions & Core Differentiators

**The Solution:** NutriSense is an AI-powered nutrition and metabolic health assistant designed specifically for Pakistani and South Asian users. Using Google Gemini AI, it calculates personalized macro targets based on medical conditions and recommends tailored workout and exercise strategies matched to each user's needs – such as strength training and calorie-dense meal plans for underweight individuals, and safe, low-impact routines for diabetics and hypertension patients. It provides a clinically-guarded AI coach that detects high-risk patterns (like hypoglycemia) and escalates to care. It includes a Ramadan fasting mode with Sehri/Iftar schedules and split hydration, multi-family profile management, bilingual English/Urdu support, offline-first sync, and weekly PDF health reports. NutriSense makes personalized, culturally-accurate nutrition and fitness guidance accessible to every Pakistani household – regardless of income.

# 2. System Architecture & Tech Stack Specification

NutriSense adopts a modern, decoupled, multi-tier client-server architecture designed for high availability, sub-second latency, and enterprise security compliance.

## 2.1 High-Level Layered Architecture Overview

The system is organized into five distinct architectural tiers:

1. **Presentation Layer (Client):** Multi-platform Flutter application providing responsive, high-contrast dark-mode UI with bilingual Urdu/English localization.
2. **State & Service Controller Layer:** Reactive ListenableBuilder controllers, Singleton ViewModels, and background sync workers managing local business logic.
3. **Backend API Gateway Layer:** Asynchronous FastAPI microservices providing secure REST endpoints, input sanitization, and structured JSON schemas.
4. **AI & Intelligence Layer:** Google Gemini LLMs governed by a multi-key failover pool and clinical safety guardrail engines.
5. **Data Persistence Layer:** Supabase PostgreSQL with Row Level Security (RLS) policies coupled with an offline-first mobile SQLite caching layer.

## 2.2 Frontend Architecture & Client-Side Stack

The mobile and web client is engineered using Flutter 3.x, prioritizing native performance, modularity, and accessibility.

| Component Layer | Technology / Library | Architectural Purpose & Scope |
|---|---|---|
| Framework | Flutter 3.x / Dart 3.x | Cross-platform high-performance rendering engine (Android, iOS, Web). |
| State Management | Provider & ListenableBuilder | Lightweight, reactive state propagation for ViewModels and controllers. |
| Local Database | sqflite / SQLite | Offline-first transactional local cache for meal logs, water logs, and habits. |
| Preferences | shared_preferences | Secure local persistence for tokens, language, and Ramadan state flags. |
| Health Integration | health package (v12+) | Bi-directional bridge to Google Health Connect and Apple HealthKit. |
| Voice & Speech | flutter_tts | Bilingual Text-To-Speech audio synthesizer for English and Urdu coaching. |
| Networking | http & connectivity_plus | REST API communication with dynamic network status monitoring. |

## 2.3 Backend Architecture & Server-Side Stack

The backend infrastructure is built on FastAPI (Python 3.11+ ASGI), chosen for native asynchronous I/O and rapid JSON serialization.

| Service Module | Core Technology | Operational Capability & Responsibilities |
|---|---|---|
| API Server | FastAPI / Uvicorn (ASGI) | Asynchronous endpoints handling meal search, coaching, and sync. |
| Cloud Database | Supabase PostgreSQL | Relational database with strict Row Level Security (RLS) and Auth. |
| Authentication | Supabase Auth (JWT) | Encrypted JSON Web Tokens with bearer verification middleware. |
| PDF Reporting | ReportLab Engine | Generates high-resolution weekly clinical receipts and dietary summaries. |
| HTTP Client | httpx (Async) | High-throughput asynchronous communication with external third-party APIs. |

### 2.4 AI Engine & Multi-Key Failover Pool Architecture

NutriSense leverages Google's flagship Gemini models (gemini-2.5-flash and gemini-3.6-flash). To ensure 99.9% uptime and prevent service denial from Google API rate limits, a custom GeminiPool manager executes automatic round-robin load distribution and instant failover.

> **Resilient AI Pool Design Pattern**
> When an inference call triggers an HTTP 429 Quota Exceeded error, GeminiPool catches the exception, masks sensitive credentials in logs, instantly rotates the active index to the next verified key, and transparently completes inference in <500ms without user interruption.

### 2.5 External Integrations & Third-Party APIs

- **OpenFoodFacts API:** Global barcode lookup providing raw ingredient and nutritional metadata for packaged goods.
- **OpenStreetMap / Overpass API:** Geospatial query engine discovering certified emergency hospitals, clinics, and pharmacies within a 5km to 15km user radius.
- **Aladhan Prayer API:** Astronomical calculation engine providing exact daily Fajr (Sehri) and Maghrib (Iftar) timestamps based on GPS coordinates.

## 3. Software Engineering Diagrams & Specifications

Here are eight foundational architectural and behavioural diagrams specifying NutriSense's software design. Each specification outlines the actors, component interactions, lifecycle states, and data pathways.

## 3.1 Figure 01 : High-Level System Architecture Diagram

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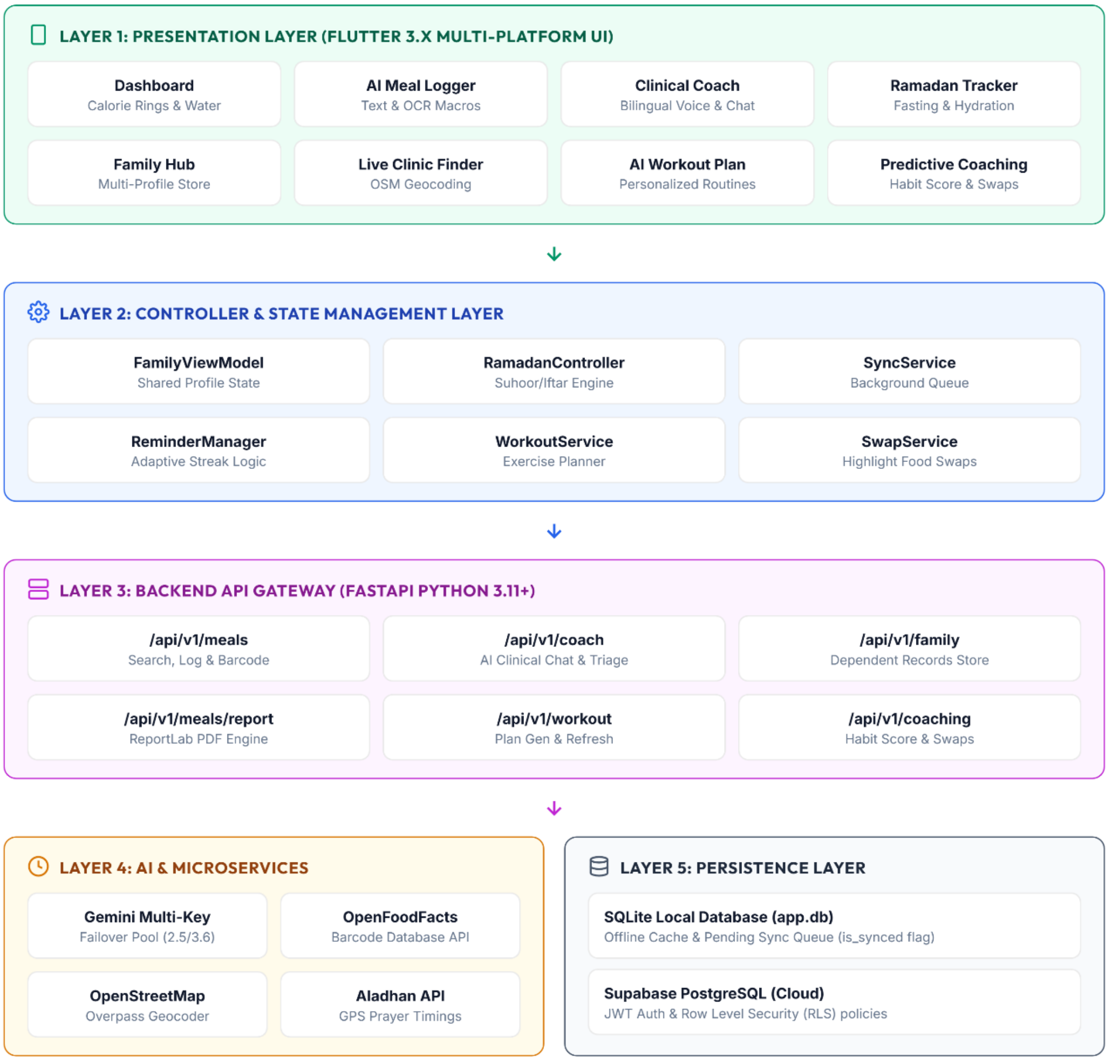


- **Architectural Walkthrough**: The Presentation Layer communicates with the Controller Layer via reactive state notifications. The Controller Layer routes network requests through the ApiClient to FastAPI endpoints. FastAPI orchestrates Gemini AI inference and third-party APIs, persisting validated data to Supabase while the client maintains an offline-first SQLite mirror.

## 3.2 Figure 02 : Use Case Diagram

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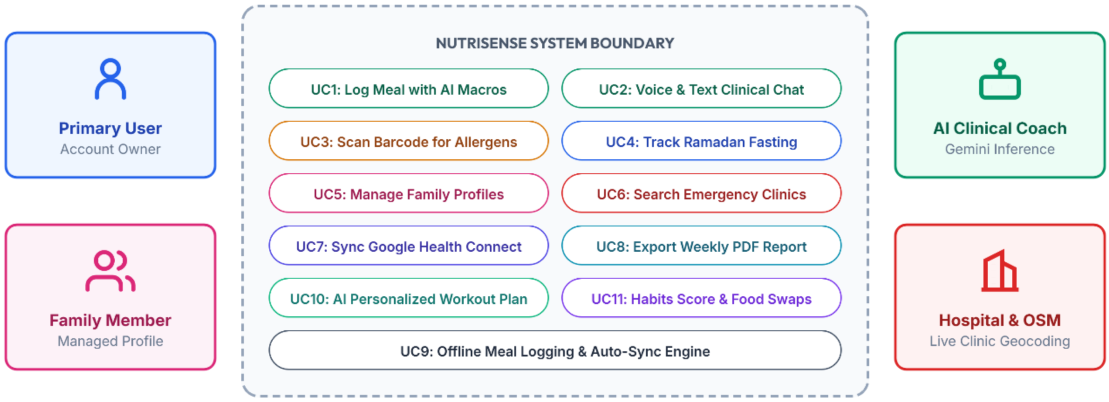


- **Use Case Specifications:** The Primary User initiates core workflows (Meal Logging, Fasting Mode, Family Management). The AI Coach acts as an active actor providing proactive dietary recommendations and clinical symptom triage. External Emergency Services receive geolocation requests to provide direct turn-by-turn routing.

## 3.3 Figure 03 : Entity-Relationship Diagram

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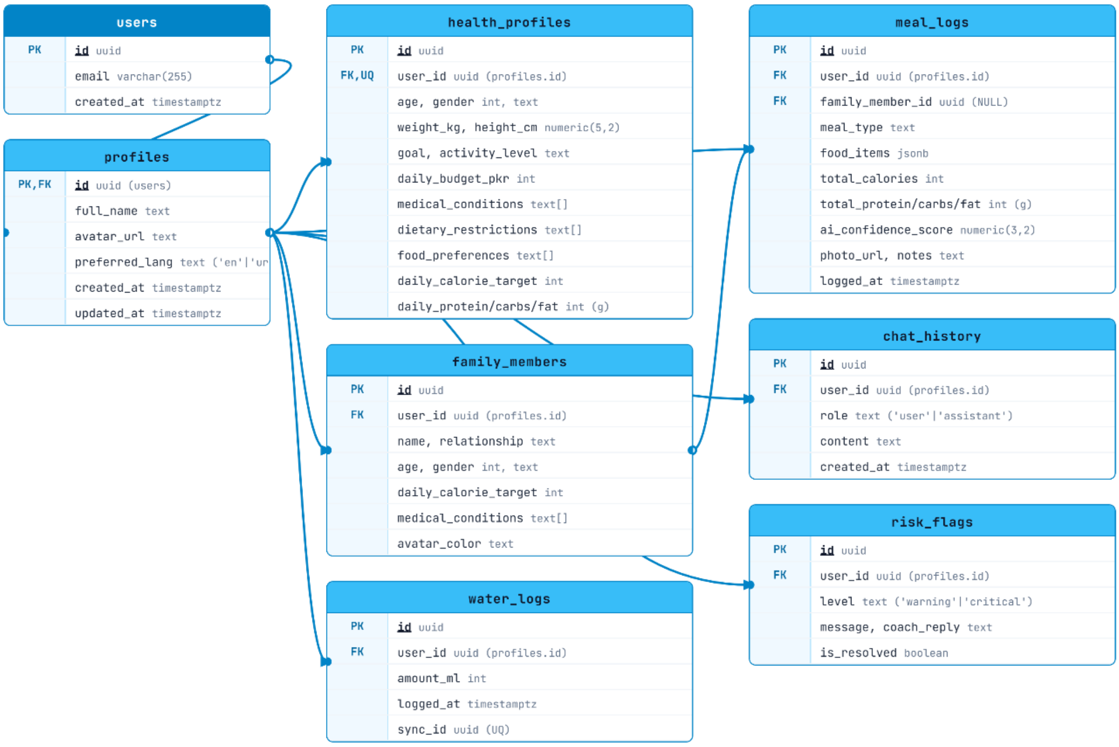


The database schema is normalized to 3NF, enforcing referential integrity and user isolation via foreign key cascades:

| Table Name | Key Columns & Types | Relational Constraints & Description |
|---|---|---|
| users | id (UUID PK), email, created_at | Root Supabase Auth entity representing registered accounts. |
| health_profiles | user_id (UUID FK), age, height, weight, goal, conditions[], allergies[] | 1:1 with users. Stores clinical metrics, daily targets, and allergies. |
| family_members | id (UUID PK), primary_user_id (FK), name, relationship, age, conditions[] | 1:N with users. Stores dependent sub-profiles for family tracking. |
| meal_logs | id (UUID PK), user_id (FK), family_member_id (FK), notes, calories, macros | Stores logged meals with macro breakdown and offline sync flags. |
| chat_history | id (UUID PK), user_id (FK), sender, message, metadata, timestamp | Maintains multi-turn conversations between user and AI Health Coach. |

| | | |
|---|---|---|
| water_logs | id (UUID PK), user_id (FK), amount_ml, logged_at | Logs hydration entries for daily water target tracking. |
| habit_scores | id (UUID PK), user_id (FK), score, streak_days, last_calculated | Maintains 30-day algorithmic consistency scores and streaks. |

## 3.4 Figure 04 : AI Meal Logging & Macro Estimation Flowchart

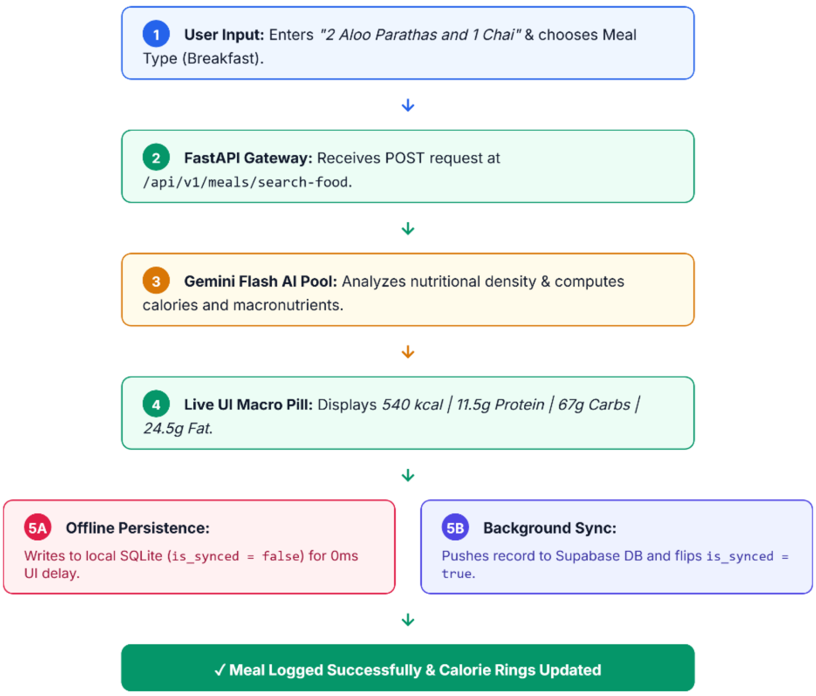

Figure 04 : AI Meal Logging & Macro Estimation Flowchart

- **Lifecycle Execution:**

**(1)** User inputs '1 plate Chicken Biryani with Raita'.

**(2)** Flutter sends JSON payload to FastAPI `/meals/search-food`.

**(3)** FastAPI prompts GeminiPool with USDA and South Asian culinary groundings.

**(4)** Gemini returns structured JSON: 520 kcal, 28g protein, 65g carbs, 16g fat.

**(5)** UI displays macro summary for 1-tap confirmation.

**(6)** On confirmation, row is written to local SQLite and queued for background Supabase push.

## 3.5 Figure 05 : Gemini Multi-Key Failover Flowchart

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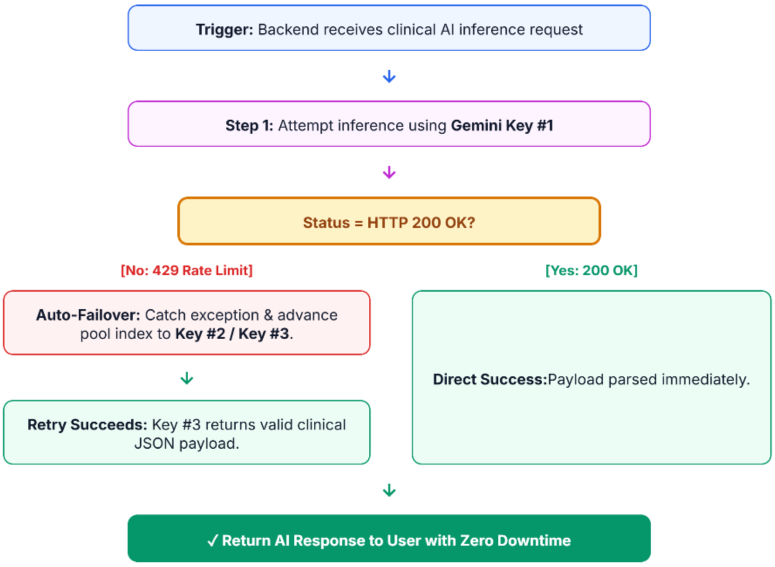


- **Failover Mechanism:** When key quota is exhausted, the pool catches `google.genai.errors.ClientError (429)`, logs a warning with masked key identifiers, increments the active pointer, and re-executes inference with zero dropped packets.

## 3.6 Figure 06 : Software Component Diagram

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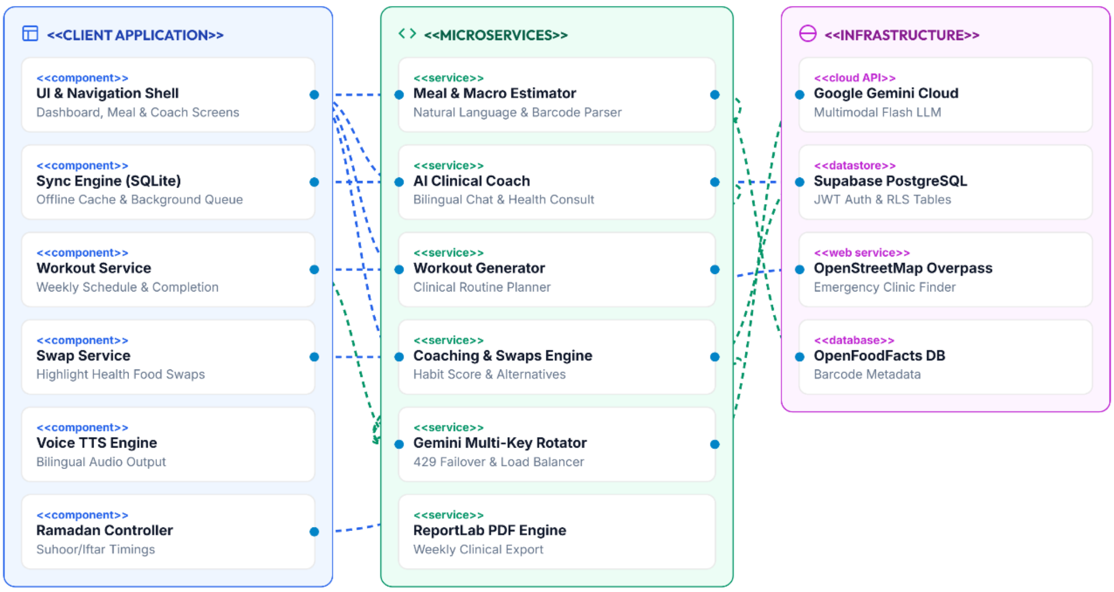
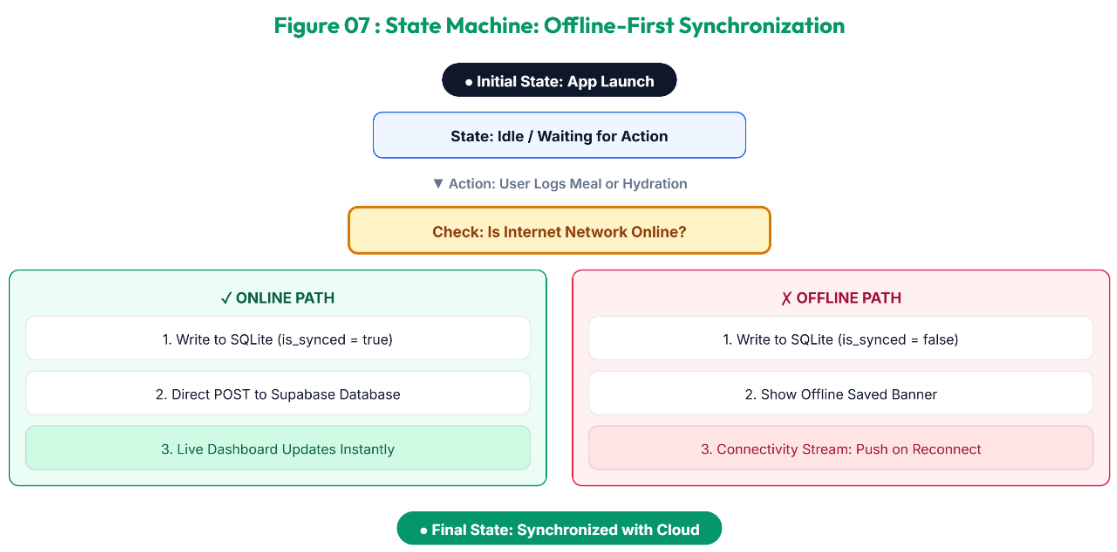


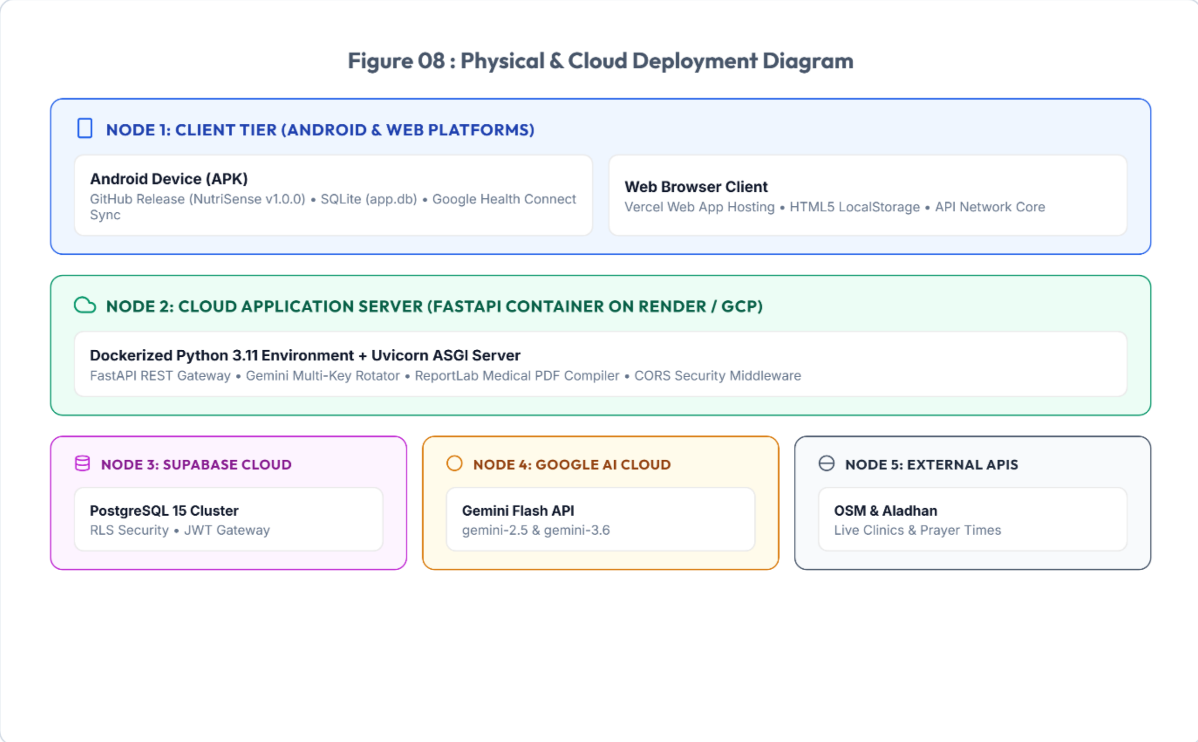
- **Component Modularity:** Ensures strict decoupling. The UI layer communicates solely through high-level ViewModels and Service singletons, allowing backend endpoints or AI models to be swapped without frontend refactoring.

## 3.7 Figure 07 : State Machine: Offline-First Synchronization



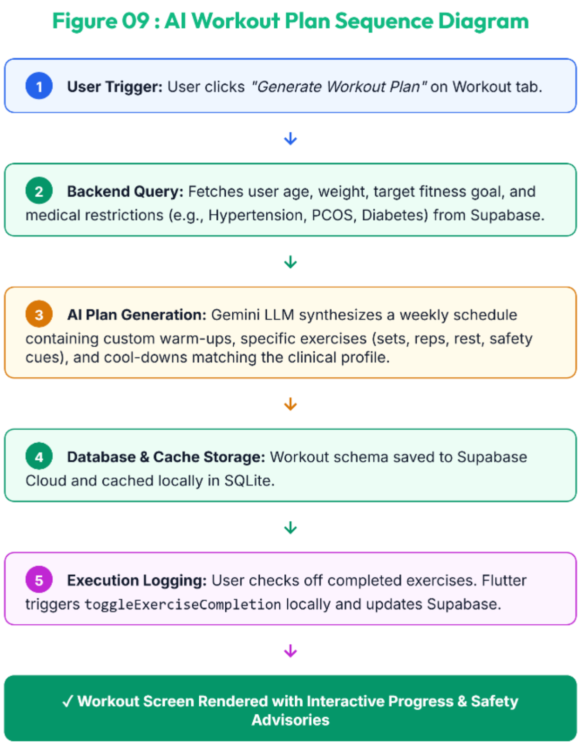

- **Data Synchronization Guarantee:** The state machine guarantees zero data loss during network dropouts by treating local SQLite as the definitive client source of truth.

## 3.8 Figure 08 : Physical & Cloud Deployment Diagram




- **Infrastructure Resilience:** Employs TLS 1.3 encryption end-to-end, containerized stateless FastAPI application nodes, and managed Supabase PostgreSQL with automated daily backups.

## 3.9 Figure 09 : AI Workout Plan Sequence Diagram




- **Plan Generation Lifecycle:**

**(1)** User requests a plan.

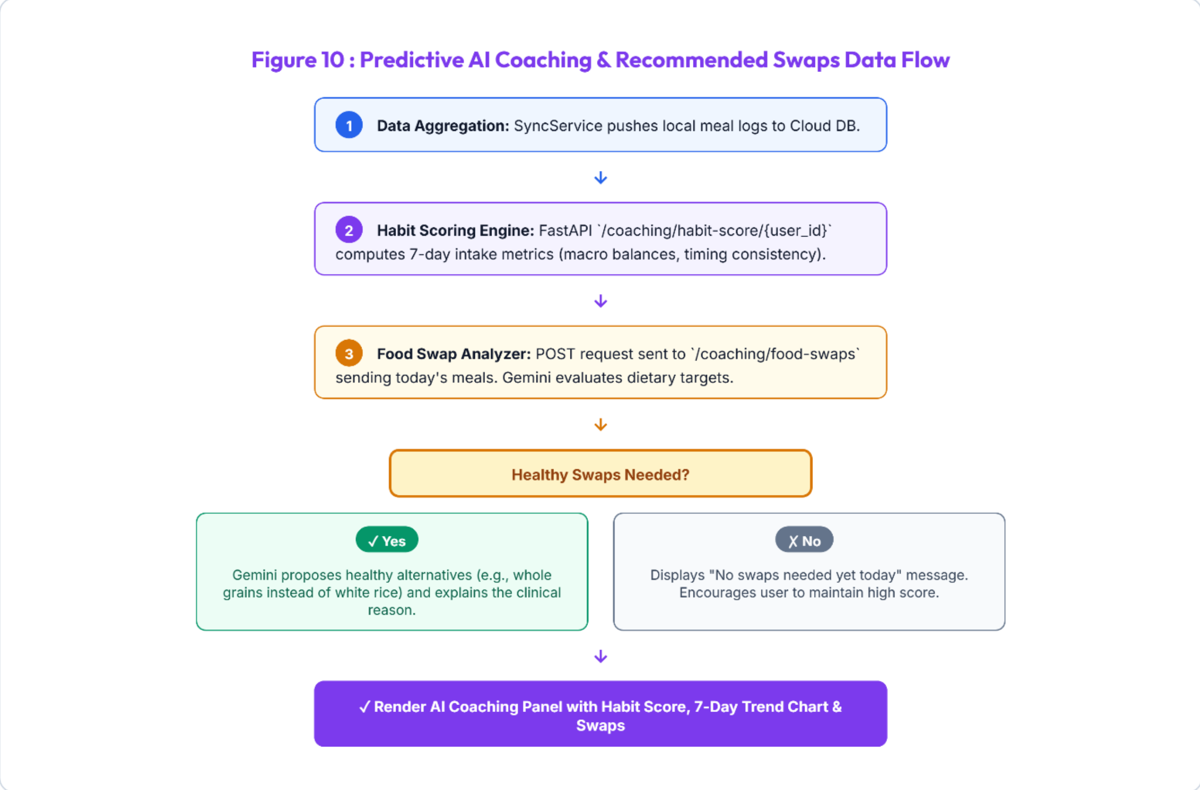
**(2)** Backend queries profile metrics and medical restrictions from Supabase.

**(3)** Gemini AI synthesizes a weekly schedule containing custom warm-ups, specific exercises (sets, reps, rest, safety cues), and cool-downs matching the clinical profile.

**(4)** Workout plan is saved to Supabase and cached locally in SQLite.

**(5)** User checks off completed exercises, triggering toggleExerciseCompletion locally and updating Supabase.

## 3.10 Figure 10 : Predictive AI Coaching & Recommended Swaps Data Flow




- **Optimization Processing Loop:**

**(1)** SyncService pushes meal logs to Cloud DB.

**(2)** Habit Scoring Engine calculates 7-day habit scores.

**(3)** Food Swap Analyzer queries Gemini for today's diet swaps.

**(4)** If swaps are needed, Gemini proposes healthy alternatives and clinical reasons.

**(5)** AI Coaching Panel renders in UI with habit score, trend chart, and swaps.

# 4. Core Functional Module Specifications

## 4.1 Module 1: Natural Language AI Auto-Macro Estimator

The AI Auto-Macro Estimator eliminates the complexity of calorie counting. Users describe their meals in natural conversational language in English or Urdu without typing numbers.

- **Bilingual Natural Language Parsing:** Accurately understands colloquial meal names (e.g., '2 Aloo Parathas and 1 Chai', 'Daal Chawal with salad', 'Grilled Salmon with quinoa').
- **Portion Intelligence:** Automatically determines standard serving sizes when quantities are omitted, or scales macros dynamically based on specified quantities.
- **Regional & Desi Cuisine Grounding:** Built on an extensive South Asian nutritional knowledge matrix accounting for ghee, oil, and traditional cooking methods.

## 4.2 Module 2: High-Availability Multi-Key Gemini Failover Pool

To deliver enterprise SLA standards, the `GeminiPool` singleton implements thread-safe key management:

- **Dynamic Round-Robin Rotation:** Distributes API load across multiple Gemini API keys to balance quota consumption.
- **Instant HTTP 429 Interception:** Intercepts rate-limiting responses and switches keys in <50ms.
- **Security Redaction:** Automatically masks API key strings in server logs to prevent credential leakage.

## 4.3 Module 3: Multi-Member Family & Dependent Health Profiles

NutriSense allows a primary account holder to manage the nutritional wellness of their entire household:

- **Dependent Profiles**: Supports adding children, elderly parents, spouses, and siblings with custom age, gender, and calorie targets.
- **Allergy & Medical Rules per Member:** Enforces isolated safety guardrails (e.g., flagging peanut allergens for Child A while tracking low-sodium for Parent B).
- **1-Tap Dashboard Context Switching:** Allows switching the primary dashboard to visualize meals and calorie progress for any family member instantly.

## 4.4 Module 4: Ramadan Fasting Intelligence & Hydration Engine

A dedicated operational mode tailoring the entire app experience for religious fasting observances:

- **Solar Calculation Engine:** Automatically fetches astronomical Sehri (Fajr) and Iftar (Maghrib) timings based on exact GPS coordinates.
- **Live Fasting Countdown:** Displays hours and minutes remaining until Iftar / Sehri with an animated circular progress indicator.
- **Targeted Hydration Presets:** Provides 1-tap logging chips tailored for non-fasting windows (+500ml Iftar, +250ml Taraweeh, +500ml Sehri).
- **Dynamic Meal Categorization:** Replaces standard Breakfast/Lunch slots with Sehri, Iftar, and Post-Taraweeh Snack.

## 4.5 Module 5: Real-Time AI Clinical Chat & Bilingual Voice Coach

An interactive AI dietitian available 24/7 for conversational guidance, recipe suggestions, and health habit evaluation:

- **Rich Markdown Output:** Renders bold key terms, bulleted dietary plans, and numbered instructions cleanly.

- **Bilingual Text-To-Speech (TTS):** Reads coaching responses aloud in natural English or Urdu via `flutter_tts`.
- **Clinical Symptom Triaging:** Detects red-flag medical emergencies (chest pain, acute hypoglycemia, severe breathlessness) and directs users to immediate medical care.

## 4.6 Module 6: Emergency Clinic & Hospital Geo-Finder

Provides life-saving access to healthcare facilities during dietary or medical emergencies:

- **Overpass API Integration:** Real-time geospatial queries scan certified clinics, hospitals, and pharmacies within a 5km to 15km radius.
- **Facility Metadata:** Displays distance in kilometers, 24/7 emergency readiness flags, and direct phone contact links.
- **1-Tap Route Navigation:** Opens Google Maps or Apple Maps with pre-populated turn-by-turn emergency routing.

## 4.7 Module 7: Clinical Weekly Summary & PDF Receipt Generator

Aggregates 7-day health telemetry into professional clinical reports using ReportLab:

- **Caloric & Macronutrient Balance:** Visualizes average daily intake against prescribed medical targets.
- **Consistency & Habit Scoring:** Evaluates 30-day tracking consistency with streak metrics.
- **Doctor-Ready PDF Receipt:** Generates a downloadable clinical PDF summary suitable for sharing with physicians and dietitians.

## 4.8 Module 8: AI Workout Planner

Empowers users with personalized, clinically-safe physical training regimens tailored to their specific fitness goals, medical history, and physical profile:

- **Automated Weekly Schedule:** Generates a 7-day structured training protocol matching primary health goals (e.g., strength training for underweight, low-impact cardio for hypertension/diabetics).
- **Interactive Exercise Runtimes:** Tracks exercise execution and completion status in real-time, allowing users to toggle completion status on active screens.
- **Form Cues & Clinical Precautions:** Integrates precise training safety alerts and form cues generated dynamically by Gemini to prevent injuries.

## 4.9 Module 9: Predictive AI Coaching & Food Swaps

Provides proactive metabolic health optimization by analyzing rolling habits and delivering smart alternatives for high-glycemic or unhealthy logged items:

- **7-Day Consistency Habit Ring:** Computes a rolling consistency score based on dietary adherence, macronutrient balances, and logging streaks.
- **Gemini Food Swap Recommendations:** Automatically flags high-glycemic index foods or dietary violations and suggests healthy alternatives (e.g., swapping white rice for brown rice).
- **Clinical Rationale Explanations:** Accompanies every swap recommendation with a clear clinical reason detailing how the swap improves metabolic markers (e.g., reducing glycemic load).

# 5. Security, Compliance & Non-Functional Requirements

## 5.1 Authentication, Authorization & Session Security

- **JWT Bearer Tokens:** All client-server communications require cryptographically signed JSON Web Tokens issued by Supabase Auth.
- **Secure Local Storage:** Sensitive tokens and credentials are stored using hardware-backed platform keystores.
- **Safe Account Termination:** Comprehensive account deletion routines purge all relational database rows upon verified user confirmation.

## 5.2 Data Privacy & PostgreSQL Row-Level Security (RLS)

- **Row-Level Security Policies:** PostgreSQL RLS policies enforce that users can only SELECT, INSERT, UPDATE, or DELETE records where `user_id == auth.uid()`.
- **Family Access Isolation:** Dependent records and family meal logs are strictly restricted to the authenticated primary account holder.
- **Zero AI Data Retention:** Food scan queries and voice chats are processed ephemerally without using personal health records for public model training.

## 5.3 Performance, Scalability & Resource Optimization

- **Sub-600ms AI Response Latency:** Asynchronous FastAPI pipeline and optimized prompt tokens deliver instantaneous macro estimates.
- **60 FPS Smooth Rendering:** Flutter UI utilizes const constructors, isolated ListenableBuilder rebuild scopes, and image caching.
- **Low Power & Data Overhead:** Background sync operations execute only upon network state change or batch queue triggers.

## 5.4 Offline Resilience & Idempotent Sync Integrity

- **Zero Data Loss:** Every meal, water, and habit log is committed immediately to SQLite before initiating network transmission.
- **Idempotent Synchronization:** Background sync operations utilize unique UUID keys to prevent duplicate records upon reconnecting.